

Notes: Bertrand, Karlan, Mullainathan, Shafir, and Zinman (QJE, 2010)

What's Advertising Content Worth? Evidence from a Consumer Credit Marketing Field Experiment

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1 Big picture

- **Question.** How much does advertising content affect real consumer demand, and can those effects be priced relative to a standard economic variable such as the interest rate?
- **Setting.** The paper studies a direct-mail field experiment by a South African consumer lender. The lender sent real loan offers to 53,194 former clients, all of whom had borrowed before and therefore understood the product.
- **Core design.** The experiment independently randomized three things: the interest rate on the loan offer, the advertising content of the mailer, and the deadline for taking up the offer.
- **Advertising content treatments.** Mailers varied features such as whether a photograph appeared, whether the photo was female/male or race/gender matched, how many example loans were shown, whether the interest rate was shown, whether a loan use was suggested, whether competitor-rate comparisons were shown, whether a cell-phone raffle was mentioned, whether the lender mentioned the client's language, and whether the rate was labelled "special" or "low."
- **Main content result.** Advertising content matters. Joint tests reject the null that all content effects are zero for applications and for loans obtained. The effects are not driven by substitution away from other lenders and do not appear to worsen default selection.
- **Most important individual content results.** Showing only one example loan, not suggesting a specific loan use, and showing a female photograph significantly increase demand. The female-photo effect is concentrated among male clients.
- **Economic magnitude.** The statistically important content effects are large. For applications, one example loan, no suggested loan use, and a female photo each increase demand by roughly as much as a 200 basis point reduction in the monthly interest rate.
- **Price benchmark.** A 100 basis point reduction in the monthly interest rate raises the application probability by about 0.3 percentage points, relative to an 8.5% application rate.
- **Deadline result.** Longer deadlines strongly increase takeup. The results do not support the behavioral prediction that short deadlines help consumers overcome procrastination. In this setting, the option value of a longer deadline dominates.
- **Mechanism interpretation.** The authors argue that many content effects are more consistent with persuasion through intuitive or peripheral processing than with pure information provision.
- **Takeaway.** Seemingly noninformative advertising content can shift real demand by amounts comparable to large price changes. But predicting which particular content feature works is difficult, so context-specific field experimentation is valuable.

2 Incremental contribution of the paper

2.1 Contribution relative to economics of advertising

- The classic economics view treats advertising mainly as information: it tells consumers about prices, availability, or quality.
- This paper brings persuasive and psychological advertising content into a clean economic field experiment.
- The experiment is not about advertising intensity, media exposure, or aggregate ad spending. It isolates the content of the same direct-mail offer.
- Because content and price are randomized simultaneously, the paper can put content effects in price-equivalent units.

Interpretation: The paper’s key conceptual contribution is to make “creative content” measurable in the same units as price. This is what allows statements such as: one example loan or a female photograph can shift demand as much as a 25% reduction in the monthly interest rate.

2.2 Contribution relative to psychology and marketing lab evidence

- Psychology and marketing studies had shown that framing, cues, complexity, affect, and salience can affect choices in laboratories.
- The paper tests related ideas in a real market with real borrowing decisions and nontrivial financial stakes.
- The sample consists of experienced consumers, not student subjects or hypothetical shoppers.
- The paper finds that some lab-motivated channels survive in the field, especially choice overload and affective response, but many predicted effects do not.

Interpretation: The incremental contribution is not simply that psychology matters. The paper shows that psychological content can matter in real financial decisions, but it also shows that external validity is highly context dependent.

2.3 Contribution relative to consumer finance

- The setting is high-interest unsecured consumer credit, where small changes in demand can have important welfare implications.
- The paper examines both takeup and repayment behavior. This matters because persuasive advertising could induce low-quality borrowing.
- The results show no clear evidence that consumers responding to content are worse risks, at least as measured by default and borrowing from other lenders.
- The deadline treatment gives evidence on procrastination, limited attention, and the option value of waiting in a credit market.

Interpretation: The paper adds to consumer finance by showing that marketing design affects credit demand directly, not only through price and underwriting terms.

2.4 Contribution relative to field-experiment methodology

- The design uses a large-scale firm field experiment with over 53,000 real offers.
- The experiment randomizes many treatments at once while maintaining independence among treatment dimensions.
- The paper demonstrates both the power and limitations of field experimentation: real decisions and large samples are valuable, but low takeup means that detecting single content effects still requires large effects.
- It makes clear that joint tests of content families can be more informative than relying exclusively on individual treatment coefficients.

Interpretation: The methodological contribution is that a firm can treat advertising content as experimentally measurable, but researchers must plan for low power when demand takeup is rare.

3 Market setting and institutional background

3.1 South African cash-loan market

- The cooperating lender operated in South Africa’s cash-loan market, serving working-poor borrowers who often lacked access to traditional bank credit.
- Loans were small, short maturity, unsecured, and high interest. The typical loan was meaningful relative to borrower income.
- Borrowers used loans for consumption smoothing and household needs such as food, clothing, education, housing, transport, and repayment of other debts.
- The market was imperfectly competitive. Borrowers did not appear to view other lenders as close substitutes, which is important for interpreting the outside-borrowing results.

Interpretation: The setting is useful because the product is familiar and financially salient. This makes it harder to explain content effects as pure information effects.

3.2 The lender’s product and relationship with borrowers

- The lender specialized in medium-maturity products, especially four-month loans.
- The modal standard monthly rate for high-risk borrowers was 11.75%, with lower standard rates for medium- and low-risk borrowers.
- The lender used a centralized credit score plus branch-level loan-officer discretion.
- Repeat borrowers had lower rejection rates than first-time applicants because the lender had repayment histories for them.
- The lender used carrots and sticks to support repayment: future access, lower prices, credit-bureau reporting, collection calls, court summons, and wage garnishment.

Interpretation: Former clients have prior product knowledge. That makes the experiment especially relevant for testing persuasive content rather than basic advertising information.

3.3 Why the product is a good test case

- The loan product is not a luxury good. Borrowing is an intermediate financial input for household needs.
- The mailer is a low-cost communication device, so small design changes are plausible firm choices.
- The outcome is observable: whether a client applies, obtains a loan, borrows from another lender, and defaults.
- Since interest rates are randomized simultaneously, the content effects can be benchmarked against a clean price effect.

Interpretation: The experiment directly asks whether seemingly superficial content changes shift consequential financial behavior.

4 Experimental design and measurement

4.1 Sample frame

- The sample consists of 53,194 former clients.
- Each client had borrowed from the lender within the previous 24 months, but not within the previous 6 months.
- The average client had taken about four previous loans.
- The mean time since the most recent loan was about 10 months.
- The full sample is close to gender balanced, with 48% female clients, and most clients are classified as African.

Interpretation: The sample is not a random sample of all households. It is a sample of experienced former borrowers whom the lender wanted to reactivate.

4.2 Treatment dimensions

The experiment varied three treatment dimensions independently:

1. **Loan price:** the monthly interest rate in the mailer.
2. **Advertising content:** eight mailer features, represented by thirteen content variables in the regressions.
3. **Deadline:** the time window for taking up the offer, roughly two, four, or six weeks, with one short-deadline extension option.

Interpretation: Independent randomization is crucial because it allows content effects to be estimated while controlling for the randomized price and deadline, and vice versa.

4.3 Interest-rate randomization

- Interest rates were randomized within preapproved risk categories.
- The standard four-month monthly rates were 7.75% for low risk, 9.75% for medium risk, and 11.75% for high risk.
- Experimental offer rates ranged from 3.25% to 11.75% per month.
- Almost all offers were discounted relative to the lender’s standard rates.
- The randomized offer rate is the price variable used to scale content effects.

Interpretation: Price randomization creates a clean demand benchmark. Without it, the authors could show that content matters, but not how much it matters relative to a meaningful economic lever.

4.4 Advertising content features

The eight content features are:

- **Photo:** no photo, male photo, female photo, race-matched photo, gender-matched photo.
- **Number of example loans:** one example loan versus several example loan amounts/maturities.
- **Interest-rate display:** whether the example table shows the interest rate or only monthly payments.

- **Suggested loan use:** no specific use versus suggestions such as school, home repair, appliance purchase, or paying off debt.
- **Competitor comparison:** no comparison, gain frame, or loss frame relative to outside rates.
- **Cell-phone raffle:** raffle mentioned or not mentioned.
- **Language affinity:** whether the mailer says the lender speaks the client's language.
- **Special/low rate label:** whether the rate is labelled as special or low.

Interpretation: The features are not all equally motivated. Some are directly linked to psychological hypotheses; others were suggested by the lender because they were easy to test.

4.5 System I and System II grouping

- The paper groups several content features by whether they plausibly trigger intuitive processing or deliberative processing.
- **System I** features include photo, number of example loans, and interest-rate display. These are intended to work through affect, similarity, simplicity, or peripheral cues.
- **System II** features include suggested loan use, competitor comparison, and cell-phone raffle. These are more likely to trigger explicit reasoning or justification.
- The lender-imposed language and special/low-rate labels are treated separately.

Interpretation: This thematic grouping helps interpret the results, but the paper is careful that the experiment does not sharply identify the cognitive process behind each effect.

4.6 Outcomes

The main outcomes are:

- whether the client applied before the deadline;
- whether the client obtained a loan before the deadline;
- loan amount obtained;
- whether the loan entered collection/default status;
- whether the client borrowed from another formal lender.

Interpretation: The default and outside-borrowing outcomes are important because they test whether content-generated demand is low quality or merely substitution from other lenders.

5 Empirical model and conceptual framework

5.1 Main empirical specification

The paper estimates specifications of the form

$$Y_i = f(r_i, c_{i1}, c_{i2}, \dots, c_{i13}, d_i, X_i), \quad (1)$$

where

- Y_i is a demand, borrowing, default, or outside-borrowing outcome;
- r_i is the randomized monthly interest rate;
- $c_{i1}, \dots, c_{i13}$ are advertising content treatment indicators;
- d_i is the randomized deadline treatment;
- X_i contains randomization-condition controls such as risk category, wave, and eligibility conditions.

Interpretation: Because the price, content, and deadline treatments are randomized independently, coefficients are interpretable as experimental treatment effects after conditioning on randomization strata.

5.2 Estimation choices

- Binary outcomes are estimated with probit models and reported as marginal effects.
- Loan amounts are estimated with OLS.
- Standard errors are Huber–White robust standard errors.
- The paper emphasizes joint F-tests for all content treatments because individual content effects are underpowered unless the effect is economically large.

Interpretation: The paper’s low takeup rate makes power a real issue. The sample of 53,194 is large, but detecting a tiny content effect would require hundreds of thousands of observations.

5.3 Scaling content effects by price

Let β_r be the interest-rate coefficient and β_c be a content coefficient. The price-equivalent effect is approximately

$$\text{Price-equivalent basis points} = 100 \times \frac{\beta_c}{|\beta_r|}. \quad (2)$$

Interpretation: This transforms a content effect into the interest-rate reduction that would produce a similar change in demand. This is the main reason the paper can say advertising content is economically large.

5.4 Simple benchmark without advertising effects

The baseline choice rule is

$$u_i(l) - p_i > 0, \quad (3)$$

where $u_i(l)$ is the discounted utility gain from borrowing and p_i is the cost of borrowing.

Interpretation: In a strict model where advertising content does not affect utility, beliefs, or decision weights, the content variables should have zero effects.

5.5 Informative advertising

An informative advertising model allows content to influence beliefs:

$$E_t^u(C_{it})[u_i(l)] - E_t^p(C_{it})[p_i] > 0. \quad (4)$$

Interpretation: Content may matter because it changes perceived utility or perceived price. The authors view this as less likely because the sample already knows the lender and product.

5.6 Complementary advertising

Advertising may be complementary to consumption:

$$u_i(l, l \cdot C_i) - p_i > 0. \quad (5)$$

Interpretation: This is more plausible for prestige or identity goods. It is less natural for a high-interest loan, which is mostly an intermediate financial product.

5.7 Persuasive advertising

Persuasive advertising affects the decision utility rather than the experienced utility:

$$D_i(u_i(l), C_i) - p_i > 0. \quad (6)$$

Interpretation: The authors lean toward this interpretation. Content may trigger affect, simplify decisions, change salience, manipulate reference points, or reduce/increase deliberation.

6 Identification and credibility

6.1 What identifies the effects

- Identification comes from random assignment of prices, content features, and deadlines.
- The randomizations are independent, so content effects are not confounded by prices or deadlines.
- The same lender, product, and operational environment are used for all clients.
- Outcomes are real applications, real loans, real repayment outcomes, and credit-bureau records.

Interpretation: This is a strong research design for estimating reduced-form effects of mailer content on loan demand.

6.2 Why prior borrower experience matters

- All recipients were former borrowers.
- They likely understood branch procedures, loan terms, and the lender's basic product.
- This makes basic product information less likely to explain the results.
- It makes noninformative content effects more persuasive as evidence of framing, cues, and psychological channels.

Interpretation: If new consumers had responded to content, one might argue that the mailer taught them something. With experienced borrowers, that explanation is weaker.

6.3 Threats and limitations

- The experiment is underpowered for small individual content effects.
- Multiple content treatments create multiple-testing concerns.
- The design does not fully distinguish informative, complementary, and persuasive mechanisms.
- The sample consists of prior customers of one lender in South Africa, so external validity requires caution.
- The experiment measures short-run responses to a single direct-mail campaign, not long-run welfare or repeated advertising exposure.

Interpretation: The causal effects of the randomized treatments are credible, but the precise psychological mechanism and broader generalizability are more tentative.

6.4 Why default and outside borrowing are useful checks

- A content treatment could increase demand by attracting riskier borrowers.
- A content treatment could also increase the lender's demand merely by shifting borrowers away from other lenders.
- The paper finds no significant evidence that content increases default or outside borrowing substitution.

Interpretation: These checks support the interpretation that content shifts actual demand for this lender's product, rather than simply worsening the borrower pool or reallocating borrowing across lenders.

7 Results

7.1 Price effects

- A 100 basis point reduction in the monthly interest rate raises the probability of applying before the deadline by about 0.3 percentage points.
- The mean application rate is 8.5%, so the implied takeover price elasticity is about -0.28 .
- The effect on obtaining a loan is nearly identical to the effect on applying.
- Loan amounts also respond negatively to price.
- Default rises with higher prices, consistent with adverse selection or moral hazard in response to interest rates.
- More expensive offers do not significantly increase borrowing from other formal lenders.

Interpretation: Price matters, but demand is not extremely price elastic in this sample. This makes the content effects especially striking when they are scaled by price.

7.2 Advertising content effects

- Joint tests reject the hypothesis that all content variables have zero effect on application behavior at conventional or near-conventional levels.
- The joint content p-value is about 0.07 for applying before the deadline and about 0.04 for obtaining a loan.
- Content effects are not jointly significant for loan amount, default, or outside borrowing.
- The psychology-motivated content variables are jointly significant for application and loan-obtaining outcomes; the lender-imposed content variables are not.

Interpretation: The strongest evidence is that content shifts the extensive margin: whether people apply and whether they obtain a loan.

7.3 Important individual content effects

- **Female photo.** A female photo raises application probability in the full sample and has a stronger effect among male clients.
- **One example loan.** Showing one example loan rather than more examples raises takeover. This supports a choice-overload or simplicity interpretation.
- **No suggested loan use.** Not suggesting a specific loan use raises takeover. The authors interpret suggested uses as potentially triggering deliberation and reinforcing the status quo of not borrowing.
- **Weak or null effects.** Race/gender matching, competitor comparisons, cell-phone raffle, language, and special/low labels generally do not show strong positive effects.

Interpretation: The effective content features are not the ones a standard informational model would emphasize. Simplicity, affect, and avoiding deliberation seem more important.

7.4 Gender heterogeneity

- Male clients respond significantly to the female photo, one example loan, and no specific loan use.
- Female clients do not show statistically significant responses to the content variables.
- Sample sizes and baseline takeup rates are similar across genders, so the split is not purely a power issue.

Interpretation: The gender split supports the idea that some content works through affective or intuitive channels, especially the female photo treatment.

7.5 Price-equivalent magnitudes

- Female photo: about 197 basis points in the full sample and 316 basis points among male clients.
- One example loan: about 234 basis points in the full sample and 396 basis points among male clients.
- No specific loan use: about 203 basis points in the full sample and 336 basis points among male clients.
- These magnitudes are large relative to the observed price sensitivity.

Interpretation: A small design decision in a direct-mail advertisement can have the demand impact of a large interest-rate cut.

7.6 Deadline effects

- Longer deadlines raise demand sharply.
- Moving from a short deadline to a medium deadline raises applications by about 3 percentage points.
- Moving to a long deadline raises applications by about 6 percentage points.
- The results do not show that short deadlines increase early applications or post-deadline applications.
- Longer deadlines also increase loans obtained and loan amounts.

Interpretation: In this setting, deadlines work like option value rather than commitment devices. Consumers value the flexibility of waiting more than the motivational benefit of urgency.

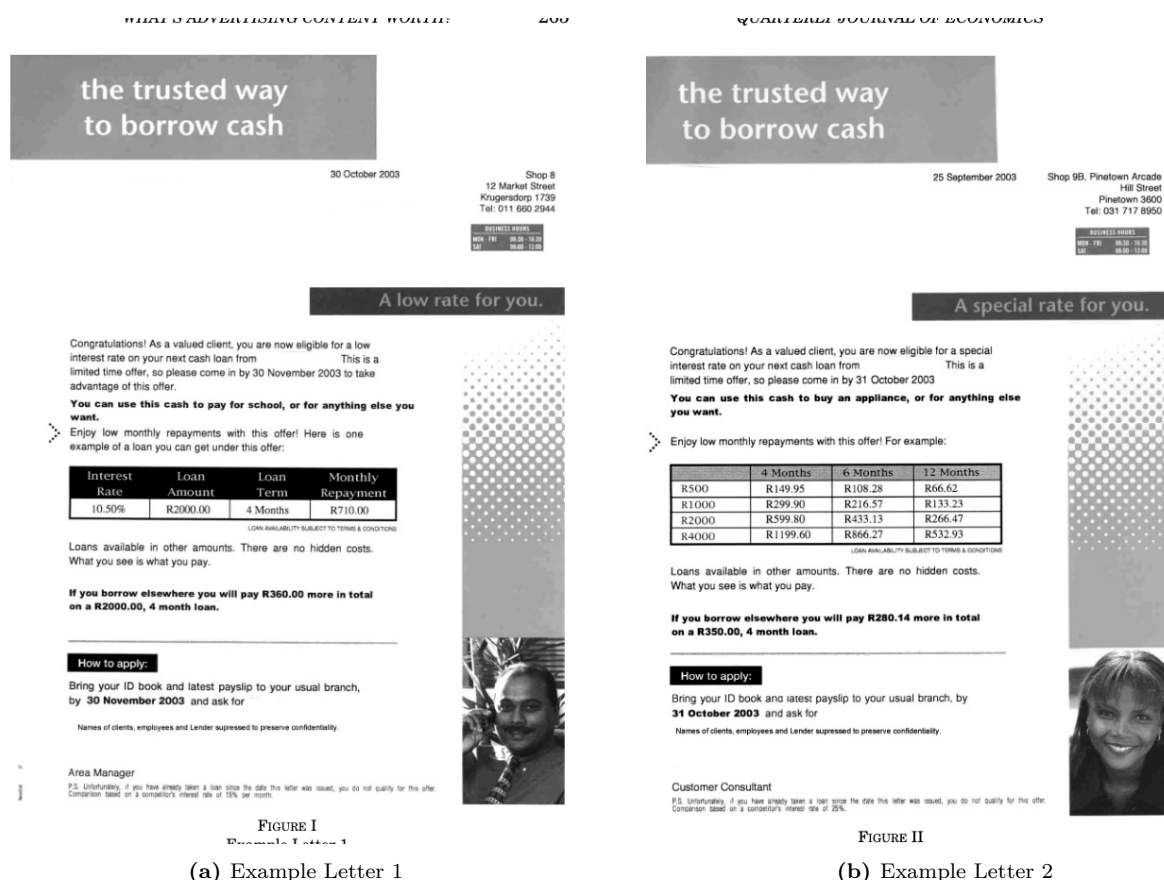
8 Tables and figures

8.1 Figures I and II: example letters

Read:

- The two sample letters show how subtle the treatments were: the basic product offer is the same, but the photo, wording, example-loan table, and comparison content can differ.
- The figures make clear that this is a direct-mail design experiment rather than a mass-media advertising experiment.
- The letters contain real terms, branch instructions, deadlines, and personalized loan offers.

Economic message: The figures illustrate that the paper is measuring the value of content variation within otherwise similar offers.



8.2 Table I: experimental summary

Read:

- Table I lists the full treatment space.
- The first group of features is classified as System I: photos, number of example loans, and interest-rate display.
- The second group is System II: suggested loan uses, competitor comparison, and cell-phone raffle.

- The lender-imposed treatments are language affinity and special/low rate wording.
- The table also reports the randomized interest-rate ranges and deadline assignment probabilities.

Economic message: Table I is the experimental blueprint. It shows why the paper can separately study price, advertising content, and deadline effects.

Creative content and its hypothesized effects on demand	Treatment value	Frequency	Sample frame/conditions
Feature 1: Photo	Features 1–3: System I (intuitive processing) treatments No photo	0.20	All
	Black photo	0.48	Assigned conditional on client's race to produce the targeted ratio of client-photo matches
	Non-black photo:		
	Indian	0.13	
	White	0.12	
	Colored	0.07	
Match increases due to affinity/similarity	Photo with race matched to client race	0.53	
	Photo with mismatched race	0.27	
Female increases due to affective response	Female photo	0.40	
	Male photo	0.40	
Match increases due to affinity/similarity	Photo with gender matched to client gender	0.40	
	Photo with mismatched gender	0.40	
Feature 2: Number of example loans	One loan amount shown in example table	0.43	All
One loan increases: simplified choice avoids "choice overload" problem	Of low and medium risk clients	0.15	
	Of high risk clients	0.52	
	Four loan amounts shown in example table	0.57	
	Four loan amounts in table, one maturity (high risk clients)	0.48	

(a) Table I, part 1

Creative content and its hypothesized effects on demand	Treatment value	Frequency	Sample frame/conditions
	Four loan amounts in table, one maturity (low/med risk clients)	0.75	
	Four loan amounts in table, three maturities (low/med risk clients)	0.10	Only low and medium risk eligible for 4 amount, 3 maturity treatment
Feature 3: Interest rate shown in example(s)?	Interest rate shown (and monthly payments)	0.80	All
Indeterminate: several potentially counteracting channels (see Section III.F of text for details)	Interest rate not shown (just monthly payments)	0.20	
Feature 4: Suggested loan uses	Features 4–6: System II (deliberative processing) treatments "You can use this loan for anything you want"	0.20	All
No suggested use maximizes demand, because suggesting particular uses triggers deliberation and reinforces the status quo (not borrowing)	"You can use this loan to X, or for anything else you want", where X is: Pay off a more expensive debt Buy an appliance Pay for school Repair your home	0.20 0.20 0.20 0.20	

(b) Table I, part 2

Creative content and its hypothesized effects on demand	Treatment value	Frequency	Sample frame/conditions
Comparison increases by inducing choice of dominating (Lender's) option	Feature 5: Comparison to outside rate No comparison to competitor rates	0.20	All
Loss frame increases by triggering loss aversion	Gain frame Loss frame	0.40 0.40	
Indeterminate: mentioning increases if overestimate small probabilities, but decreases if reason-based choice and can't justify irrelevant good	Feature 6: Cell phone raffle Mentioned cell phone raffle	0.25	All
	Not mentioned cell phone raffle	0.75	
Feature 7: Client's language	Features 7 and 8: Lender-imposed treatments No mention of language "We speak (client's language)"	0.37 0.63	Eligible if non-English primary language (0.44 of full sample)
Feature 8: "A 'special' or 'low' rate for you"	Interest rate is labeled as "special" or "low"	0.75	All

(a) Table I, part 3

Creative content and its hypothesized effects on demand	Treatment value	Frequency	Sample frame/conditions
	No mention of "special" or "low"	0.25	
Interest rate	Other treatments High risk: [3.25, 11.75] Medium risk: [3.25, 9.75] Low risk: [3.25, 7.75]		Monthly rates randomly assigned from a smooth distribution, conditional on risk
Deadline	Medium deadline (approx. 4 weeks) Long deadline (approx. 6 weeks)	0.78 0.14	1.0 of sample eligible for medium 0.79 of sample eligible for long (certain branches excluded by Lender)
	Short deadline (approx. 2 weeks)	0.03	0.14 of sample eligible for short (certain branches excluded by Lender, and all P.O. boxes excluded)
	Short deadline with option to extend 2 weeks by calling in	0.04	

(b) Table I, part 4

8.3 Table II: summary statistics

Read:

- 8.5% of the full sample applied before the deadline, and 7.4% obtained a loan.
- The average unconditional loan amount is 110 rand; among borrowers it is 1,489 rand.
- 12% of obtained loans are in default/collection status.
- 22% of the full sample borrowed from another lender and did not apply with this lender.
- The mean offer rate is 7.93% per month.
- Three-quarters of the sample is high risk; 48% are female; the average number of previous loans is 4.14.

Economic message: The low application rate explains why individual content tests are underpowered unless the effects are large.

TABLE II
SUMMARY STATISTICS (MEANS OR PROPORTIONS, WITH STANDARD DEVIATIONS
IN PARENTHESES)

	Full sample	Obtained a loan	Did not obtain a loan
Applied before deadline	0.085	1	0.01
Obtained a loan before deadline	0.074	1	0
Loan amount in rand	110 (536)	1,489 (1,351)	0 (0)
Loan in default		0.12	
Got outside loan and did not apply with Lender	0.22	0.00	0.24
Maturity = 4 months		0.81	
Offer rate	7.93	7.23	7.98
Last loan amount in rand	1,118 (829)	1,158 (835)	1,115 (828)
Last maturity = 4 months	0.93	0.91	0.93
Low risk	0.14	0.30	0.12
Medium risk	0.10	0.21	0.10
High risk	0.76	0.50	0.78
Female	0.48	0.49	0.48
Predicted education (years)	6.85 (3.25)	7.08 (3.30)	6.83 (3.25)
Number previous loans with Lender	4.14 (3.77)	4.71 (4.09)	4.10 (3.74)
Months since most recent loan with Lender	10.4 (6.80)	6.19 (5.81)	10.8 (6.76)
Race = African	0.85	0.85	0.85
Race = Indian	0.03	0.03	0.03
Race = White	0.08	0.08	0.08
Race = Mixed ("Colored")	0.03	0.04	0.03
Gross monthly income in rand	3,416 (19,657)	3,424 (2,134)	3,416 (20,420)
Number of observations	53,194	3,944	49,250

8.4 Table III: effects of advertising content on borrower behavior

Read:

- The offer interest-rate coefficient is strongly negative for applications and loans obtained.
- The female-photo treatment raises application probability in the full sample and especially for male clients.
- Showing one example loan raises application and loan-obtaining probabilities.
- No specific loan use raises application probability.
- Content effects are jointly significant or near significant for takeup and loan obtained, but not for default or outside borrowing.
- The psychology-motivated content variables are jointly more important than the lender-imposed content variables.

Economic message: Table III is the main evidence that non-price advertising content shifts real demand.

Dependent variable	Applied for loan before mailer deadline	Applied for loan before mailer deadline	Applied for loan before mailer deadline	Obtained loan before mailer deadline	Loan amount obtained before mailer deadline	Loan in collection status	Borrowed from other lender
Sample Estimator	Full Probit	Males Probit	Females Probit	Full Probit	Full OLS	Obtained Probit	Full Probit
Mean (dependent variable)	0.0850	0.0824	0.0879	0.0741	110.4363	0.1207	0.2183
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Monthly interest rate in percentage point units (e.g., 8.2)	-0.0029*** (0.0005)	-0.0025*** (0.0007)	-0.0034*** (0.0008)	-0.0036*** (0.0005)	-4.7712*** (0.8238)	0.0071*** (0.0022)	0.0009 (0.0008)
1 = no photo	0.0013 (0.0040)	-0.0050 (0.0048)	0.0021 (0.0055)	0.0029 (0.0037)	3.8316 (7.6763)	0.0013 (0.0166)	-0.0024 (0.0060)
1 = female photo (System I: affective response)	0.0057** (0.0026)	0.0079** (0.0034)	0.0032 (0.0038)	0.0056** (0.0024)	8.3292 (5.0897)	-0.0076 (0.0107)	-0.0047 (0.0040)
1 = photo gender matches client's (System I: affinity/similarity)	-0.0026 (0.0026)	-0.0034 (0.0034)	-0.0038 (0.0038)	-0.0035 (0.0024)	-7.1773 (5.0850)	-0.0059 (0.0107)	0.0041 (0.0040)
1 = photo race matches client's (System I: affinity/similarity)	-0.0056 (0.0048)	-0.0014 (0.0064)	-0.0099 (0.0070)	-0.0035 (0.0044)	9.9638 (10.4079)	0.0181 (0.0176)	-0.0018 (0.0072)
1 = one example loan shown (System I: avoid choice overload)	0.0068** (0.0028)	0.0099*** (0.0038)	0.0031 (0.0040)	0.0075*** (0.0026)	2.4394 (4.8383)	0.0073 (0.0117)	-0.0043 (0.0042)
1 = interest rate shown (System I: several, potentially offsetting, channels)	0.0025 (0.0030)	-0.0017 (0.0042)	0.0073 (0.0044)	0.0043 (0.0028)	2.8879 (6.7231)	0.0140 (0.0123)	0.0007 (0.0049)

(a) Table III, part 1

Dependent variable	Applied for loan before mailer deadline	Applied for loan before mailer deadline	Applied for loan before mailer deadline	Obtained loan before mailer deadline	Loan amount obtained before mailer deadline	Loan in collection status	Borrowed from other lender
Sample Estimator	Full Probit	Males Probit	Females Probit	Full Probit	Full OLS	Obtained Probit	Full Probit
Mean (dependent variable)	0.0850	0.0824	0.0879	0.0741	110.4363	0.1207	0.2183
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
1 = cell phone raffle mentioned (System II: overestimate small probabilities vs. conflict from reason-based choice)	-0.0023 (0.0026)	-0.0001 (0.0036)	-0.0049 (0.0039)	-0.0013 (0.0025)	-9.4384* (5.1200)	-0.0050 (0.0109)	-0.0015 (0.0041)
1 = no specific loan use mentioned (System II: mentioning specific use, via text, triggers deliberation)	0.0059** (0.0029)	0.0084** (0.0040)	0.0031 (0.0043)	0.0043 (0.0027)	4.0850 (5.6266)	0.0086 (0.0121)	-0.0033 (0.0045)
1 = comparison to competitor rate (System II: makes dominating option salient)	-0.0002 (0.0031)	-0.0012 (0.0043)	0.0010 (0.0046)	0.0012 (0.0029)	-2.6021 (6.2961)	-0.0027 (0.0133)	0.0054 (0.0049)
1 = loss frame comparison (System II: triggers loss aversion)	-0.0024 (0.0026)	-0.0018 (0.0035)	-0.0029 (0.0038)	-0.0021 (0.0024)	3.0925 (5.0678)	0.0032 (0.0108)	0.0027 (0.0040)
1 = we speak "your language" (Lender imposed)	-0.0043 (0.0036)	-0.0016 (0.0049)	-0.0073 (0.0053)	-0.0036 (0.0035)	-11.3556* (6.2935)	-0.0031 (0.0152)	0.0133** (0.0059)
1 = a "low" or "special" rate for you (Lender imposed)	0.0001 (0.0031)	-0.0022 (0.0043)	0.0027 (0.0045)	0.0010 (0.0028)	3.3864 (5.9209)	-0.0137 (0.0128)	-0.0002 (0.0047)
N	53,194	27,848	25,346	53,194	53,194	3,944	53,194

(b) Table III, part 2

Dependent variable	Applied for loan before mailer deadline	Applied for loan before mailer deadline	Applied for loan before mailer deadline	Obtained loan before mailer deadline	Loan amount obtained before mailer deadline	Loan in collection status	Borrowed from other lender
Sample Estimator	Full Probit	Males Probit	Females Probit	Full Probit	Full OLS	Obtained Probit	Full Probit
Mean (dependent variable)	0.0850	0.0824	0.0879	0.0741	110.4363	0.1207	0.2183
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(Pseudo-) r^2 -squared	.0456	.0481	.0438	.0534	.0361	.0674	.0048
p -Value, F -test on all advertising content variables	.0729	.0623	.5354	.0431	.2483	.7485	.4866
Absolute value lower bound of range of joint content effect for which F -test rejects null	0.0010	0.0021		0.0026			
Absolute value upper bound of range of joint content effect for which F -test rejects null	0.0448	0.0388		0.0498			
p -Value, F -test on Lender-imposed content ("low" or "special" language)	.5064	.8217	.3337	.5254	.1695	.5382	.0785
p -Value, F -test on psychology-motivated content (all other features)	.0522	.0300	.5541	.0286	.3420	.7262	.7583

(a) Table III, part 3

Dependent variable	Applied for loan before mailer deadline	Applied for loan before mailer deadline	Applied for loan before mailer deadline	Obtained loan before mailer deadline	Loan amount obtained before mailer deadline	Loan in collection status	Borrowed from other lender
Sample Estimator	Full Probit	Males Probit	Females Probit	Full Probit	Full OLS	Obtained Probit	Full Probit
Mean (dependent variable)	0.0850	0.0824	0.0879	0.0741	110.4363	0.1207	0.2183
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Split psychology-motivated content:							
p -Value, F -test on System II (reasoning) content (suggested use, comparison, cell)	.1946	.2643	.6200	.4499	.3399	.9360	.4947
p -Value, F -test on System I (intuitive) content (photo, # loans shown, rate shown)	.0598	.0211	.3929	.0127	.4362	.4346	.7675
p -Value, F -test on System I, dropping rate shown	.0355	.0288	.5130	.0072	.3288	.4196	.7169

Note: Huber-White standard errors. Probit results are marginal effects. All models include controls for randomization conditions: risk, race, gender, language, and mailer wave (September or October). Treatment variable labels: parentheses contain summary description of our prior as to why each ad content treatment would increase demand (or of reasons) why we had no strong prior. Omitted categories: male photo, no photo gender match, no photo race match, four example loans shown, no interest rate shown, no cell phone raffle mentioned, specific loan use mentioned, no comparison to competitor rate, gain frame comparison, no mention of speaking local language, no mention of low or special rate.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

(b) Table III, notes

8.5 Table IV: content effects scaled by price effects

Read:

- The table divides each content coefficient by the price coefficient.
- Female photo is equivalent to about a 197 basis point interest-rate reduction in the full sample.
- One example loan is equivalent to about a 234 basis point reduction.
- No specific loan use is equivalent to about a 203 basis point reduction.
- The male-client price-equivalent effects are larger: roughly 316, 396, and 336 basis points for these same three variables.

Economic message: This table turns psychological or marketing content into price-equivalent units. It is the clearest evidence that advertising content can be economically large.

8.6 Table V: deadline effects

Read:

- Short deadline with extension, medium deadline, and long deadline all raise pre-deadline demand relative to the short deadline baseline.
- The long deadline raises applications by about 6 percentage points, which is very large relative to the 8.5% mean application rate.

TABLE IV
EFFECTS OF ADVERTISING CONTENT ON BORROWER BEHAVIOR: POINT ESTIMATES IN
TABLE III, SCALED BY PRICE EFFECT

Dependent variable	Applied for loan before mailer deadline	Applied for loan before mailer deadline	Applied for loan before mailer deadline
Sample	Full	Males	Females
Mean (dependent variable)	0.0850 (1)	0.0824 (2)	0.0879 (3)
1 = no photo	45	-200	62
1 = female photo (System I: affective response)	197	316	94
1 = photo gender matches client's (System I: affinity/similarity)	-90		
1 = photo race matches client's (System I: affinity/similarity)	-193	-56	-291
1 = one example loan shown (System I: avoid choice overload)	234	396	91
1 = interest rate shown (System I: several, potentially offsetting, channels)	86	-68	215
1 = cell phone raffle mentioned (System II: overestimate small probabilities vs. conflict from reason-based choice)	-79	-4	-144
1 = no specific loan use mentioned (System II: mentioning specific use, via text, triggers deliberation)	203	336	91
1 = comparison to competitor rate (System II: makes dominating option salient)	-7	-48	29
1 = loss frame comparison (System II: triggers loss aversion)	-83	-72	-85
1 = we speak "your language" (Lender imposed)	-148	-64	-215
1 = a "low" or "special" rate for you (Lender imposed)	3	-88	79

Notes: Cells divide the coefficient on the content variable from Table III by the offer rate (i.e., the price)

- Longer deadlines raise loans obtained and unconditional loan amount.
- Deadline treatments do not significantly affect default or outside borrowing.
- Panel B shows no evidence that short deadlines increase post-deadline application behavior.

Economic message: Deadline length matters more than the paper's behavioral motivation predicted. Longer deadlines appear to raise demand through option value, not by weakening procrastination.

TABLE V
EFFECTS OF DEADLINE ON BORROWER BEHAVIOR

Panel A: Predeadline demand					
Dependent variable	Applied before own deadline	Obtained loan before own deadline	Loan amount obtained before own deadline	Loan obtained before own deadline in collection status	Borrowed from other lender
Sample	Full	Full	Full	Obtained	Full
Estimator	Probit	Probit	OLS	Probit	Probit
Mean (dependent variable)	0.0850 (1)	0.0741 (2)	110.4363 (3)	0.1207 (4)	0.2183 (5)
Monthly interest rate in percentage point units (e.g., 8.2)	-0.0029*** (0.0005)	-0.0026*** (0.0005)	-4.7768*** (0.8237)	0.0075*** (0.0023)	-0.0009*** (0.0008)
Short deadline, extended	0.0322*** (0.0135)	0.0240** (0.0107)	31.1321* (17.2858)	0.0236 (0.0434)	-0.0104 (0.0131)
Medium deadline	0.0300*** (0.0068)	0.0279*** (0.0061)	38.0335*** (13.8228)	0.0205 (0.0300)	-0.0065 (0.0119)
Long deadline	0.0603*** (0.0118)	0.0563*** (0.0112)	70.1119*** (15.0945)	0.0138 (0.0363)	-0.0054 (0.0123)
(Pseudo-) <i>r</i> -squared	.0461	.0538	.0351	.0597	.0471
<i>N</i>	53,194	53,194	53,194	3,944	53,194
<i>F</i> -test of joint significance of all deadlines	0.0000	0.0000	0.0000	0.8487	0.8813

TABLE V
(CONTINUED)

Panel B: Postdeadline applications			
Dependent variable = applied	After short deadline	After medium deadline	After long deadline
Sample	Full	Full	Full
Estimator	Probit	Probit	Probit
Mean (dependent variable)	0.1830 (1)	0.1477 (2)	0.1184 (3)
Offer interest rate	-0.0010 (0.0008)	0.0005 (0.0007)	0.0009 (0.0006)
Short deadline, extended	-0.0224* (0.0117)	-0.0052 (0.0113)	-0.0030 (0.0102)
Medium deadline	-0.0058 (0.0112)	-0.0035 (0.0102)	-0.0047 (0.0092)
Long deadline	-0.0088 (0.0114)	0.0019 (0.0108)	-0.0014 (0.0095)
Pseudo <i>r</i> -squared	.0560	.0448	.0369
<i>N</i>	53,194	53,194	53,194
<i>F</i> -test of joint significance of all deadlines	0.2518	0.6332	0.8262

Notes: Huber-White standard errors. Probit results are marginal effects. All models include controls for randomization conditions: risk, mailer wave (September or October), and deadline eligibility. Short deadline is the omitted category. "Short deadline, extended" grew customers a number to call and get an extension to the medium deadline. Panel A, column (6) tests whether short deadlines spur action by inducing early applications. The dependent variable here is defined regardless of the individual's deadline length; that is, the dependent variable = 1 if the individual applied within two weeks of the mailer date, unconditional on her own deadline. Panel B: Testing three alternative measures of post-deadline takeup helps ensure that our results here are not driven by mechanical timing differences, because we have a finite number of postdeadline data (8 months). We measure postdeadline takeup using takeup after the short deadline (2 weeks), after the medium deadline (4 weeks), and after the long deadline (8 weeks). We define these outcomes for each member of the sample, regardless of their own deadline length, in order to ensure that everyone in the sample has the same takeup window. Otherwise those with the short deadline mechanically have a longer postdeadline window, and if there is a positive secular probability of raised into takeup status within the range our deadlines produce (3 to 8 months), then this would mechanically push toward a decreasing relationship between deadline length and postdeadline takeup.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

(a) Table V, Panel A: pre-deadline demand

(b) Table V, Panel B and notes

9 Short summary

- The paper studies a direct-mail field experiment in consumer credit.
- It asks how much advertising content affects real loan demand.
- The experiment randomizes price, advertising content, and deadline independently.
- The sample contains 53,194 former clients of a South African consumer lender.
- Price matters: a 100 basis point rate reduction increases applications by about 0.3 percentage points.
- Advertising content also matters: joint content effects are significant for applications and loans obtained.
- The strongest individual content treatments are female photo, one example loan, and no specific loan use.
- The demand effects of these features are roughly equivalent to large interest-rate reductions.
- Effects are stronger among male clients, especially for the female-photo treatment.
- Content effects do not appear to increase default or outside borrowing.
- Longer deadlines strongly increase demand, contrary to a procrastination-based prediction that shorter deadlines would help consumers act.
- The paper interprets the findings as evidence that persuasive and peripheral content can influence consequential choices.
- The paper also emphasizes that context matters and that ex ante prediction of which content will work is difficult.

10 Conclusion

10.1 Core conclusion

Bertrand, Karlan, Mullainathan, Shafir, and Zinman show that advertising content can have large effects on real financial decisions. In a randomized field experiment, small changes in the presentation of a loan offer significantly changed whether former borrowers applied for and obtained new loans.

10.2 Economic intuition

The intuition is that consumers do not react only to prices and product fundamentals. They also react to how choices are framed, simplified, and emotionally cued. When a mailer uses a simpler example table, avoids suggesting a specific use, or shows a female photograph, it can change the mental process by which the borrower evaluates the offer.

10.3 Incremental contribution takeaway

The paper's distinctive contribution is that it prices advertising content using simultaneous price randomization. This turns a behavioral/marketing claim into an economic magnitude: some content effects are as large as major interest-rate changes. The paper therefore bridges psychology, marketing, and economics in a field setting with real stakes.

10.4 Limitations and open questions

The experiment shows that content matters, but it does not fully identify the psychological mechanism. Informative, complementary, and persuasive channels cannot be perfectly separated. Also, the results come

from one lender, one product, one country, and one set of former clients. The main lesson is not that a specific content treatment always works, but that content effects are real, economically important, and context dependent.